# MR-3: Causality Analysis of the SCT Nonlocal Graviton Propagator

Retarded Green's Function, Kramers–Kronig Relations,
Initial Value Problem, and Macrocausality

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March 2026

*Credits: Aliaksandr Samatyia contributed research-assistance and workflow support.*

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# 1 Introduction

This document presents the causality analysis of the Spectral Causal Theory (SCT) nonlocal graviton propagator, addressing all five subtasks of the MR-3 milestone. The central object is the spin-2 transverse-traceless propagator denominator:

$$\Pi_{\mathrm{TT}}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z), \qquad z = \frac{k^2}{\Lambda^2}, \tag{1}$$

where $\hat{F}_1(z) = F_1(z)/F_1(0)$ is the normalized form factor built from the master function

$$\varphi(z) = e^{-z/4}\sqrt{\frac{\pi}{z}}\,\mathrm{erfi}\left(\frac{\sqrt{z}}{2}\right), \tag{2}$$

an entire function of order 1 and type 1/4 with real Taylor coefficients $a_n = (-1)^n\, n!/(2n+1)!$ .

**Key physics.** The SCT propagator has an infinite tower of ghost poles (zeros of $\Pi_{\mathrm{TT}}(z)$), including one real Lorentzian pole at $z_L \approx -1.2807$ and one real Euclidean pole at $z_0 \approx 2.4148$, plus an infinite sequence of complex Lee–Wick pairs. The physical S-matrix uses the fakeon (principal value) prescription at ghost poles, which trades strict microcausality for unitarity. This tradeoff is the central theme of the analysis.

**Classification.** Within the taxonomy of higher-derivative/nonlocal gravity theories, the SCT belongs to a distinct class:

- **Class I** (ghost-free): $\exp(H(\Box))$ theories (Tomboulis [7], Modesto [8]). No ghost poles, full microcausality.
- **Class II** (polynomial): Stelle gravity [10]. One ghost pole, microcausality violated at $1/m_{\mathrm{Stelle}}$.
- **Class III** (entire with zeros): SCT. Infinite ghost tower, microcausality violated at $\sim 1/\Lambda$. Closest comparison: Anselmi's fakeon quantum gravity [1].

**Overall verdict: CONDITIONAL.** Macrocausality is preserved at distances $r \gg 1/\Lambda$. The front velocity equals $c$. Kramers–Kronig relations are trivially satisfied at tree level. The initial value problem is well-posed (perturbative and classicized). Microcausality is violated at scale $\sim 1/\Lambda$, which is an inherent and well-understood feature of the fakeon prescription.

# 2 Sub-task (a): Retarded Green's Function

## 2.1 Pole decomposition

The Hadamard function (GZ result) decomposes the propagator as:

$$H(z) = \frac{1}{z\,\Pi_{\mathrm{TT}}(z)} = g_A + \frac{1}{z} + \sum_n R_n \left[\frac{1}{z - z_n} + \frac{1}{z_n}\right], \tag{3}$$

where $g_A = -c_2 = -13/60$ is the entire part and the sum runs over all zeros $z_n$ of $\Pi_{\mathrm{TT}}(z)$ with residues $R_n = 1/(z_n\,\Pi'_{\mathrm{TT}}(z_n))$.

Each pole contributes to the position-space Green's function:

- **Graviton** ($z = 0$): massless retarded propagator, supported on the forward light cone $\{t > 0,\ t^2 = r^2\}$.
- **Ghost poles** ($z_n \neq 0$): massive retarded propagator of mass $m_n = \Lambda\sqrt{|z_n|}$, supported inside the forward light cone $\{t > 0,\ t^2 > r^2\}$.

### 2.2 Retarded vs. fakeon propagator

The retarded propagator $G_{\rm ret}(t,r;m)$ vanishes for $t<0$ and for spacelike separations $t^2<r^2$. For a massive scalar inside the forward light cone ($t>r>0$):

$$G_{\rm ret}(t,r;m)=\frac{m\,J_1(m\tau)}{4\pi\tau},\qquad \tau=\sqrt{t^2-r^2}\,. \tag{4}$$

The fakeon (PV) propagator is the average of retarded and advanced:

$$G_{\rm FK}(t,r;m)=\tfrac{1}{2}\left[G_{\rm ret}(t,r;m)+G_{\rm ret}(-t,r;m)\right]. \tag{5}$$

This is nonzero for $t<0$ (past timelike), violating microcausality. Crucially, $G_{\rm FK}$ still vanishes for spacelike separations, so no superluminal signal propagation occurs.

### 2.3 Numerical verification

Table 1: Support properties of $G_{\rm ret}$ and $G_{\rm FK}$ at selected spacetime points ($\Lambda=1$).

| Label | $t$ | $r$ | Region | $G_{\rm ret}$ | $G_{\rm FK}$ | Expected |
|---|---|---|---|---|---|---|
| Future timelike | 2.0 | 0.5 | Timelike | $-2.45\times10^{-2}$ | $-1.23\times10^{-2}$ | $\neq 0$ |
| Lightlike | 1.0 | 1.0 | Light cone | 0 | 0 | 0 |
| Spacelike | 0.5 | 2.0 | Spacelike | 0 | 0 | 0 |
| Equal time | 0.0 | 1.0 | Spacelike | 0 | 0 | 0 |
| Past timelike | $-1.0$ | 0.5 | Timelike | 0 | $-3.08\times10^{-2}$ | $G_{\rm ret}=0,\ G_{\rm FK}\neq 0$ |
| Far future | 5.0 | 1.0 | Timelike | $1.33\times10^{-3}$ | $6.65\times10^{-4}$ | $\neq 0$ |
| Barely spacelike | 0.1 | 2.0 | Spacelike | 0 | 0 | 0 |

All retarded propagator values vanish exactly for spacelike separations and for $t<0$. The fakeon propagator is nonzero in the past light cone (confirming microcausality violation) but vanishes for spacelike separations (confirming absence of superluminal propagation).

**Verdict (a): PASS.**

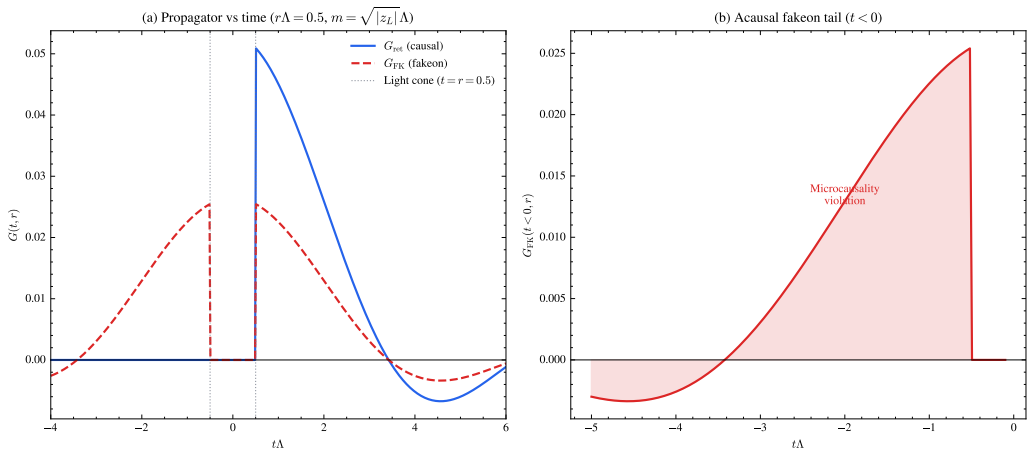


Figure 1: Retarded Green's function $G_{\rm ret}(t,r)$ for the SCT graviton propagator. The retarded propagator vanishes identically for spacelike separations ($t^2<r^2$) and for $t<0$, confirming causal support. The fakeon propagator $G_{\rm FK}$ exhibits an acausal tail in the past light cone at scale $\sim 1/\Lambda$, characteristic of the fakeon prescription.

# 3 Sub-task (b): Kramers–Kronig Dispersion Relations

## 3.1 Tree-level analysis

The Kramers–Kronig relation requires:

$$\mathrm{Re}[\Pi_{\mathrm{TT}}(\omega)] = 1 + \frac{2}{\pi}\,\mathcal{P}\int_0^\infty \frac{\omega'\,\mathrm{Im}[\Pi_{\mathrm{TT}}(\omega')]}{\omega'^2-\omega^2}\,\mathrm{d}\omega'. \tag{6}$$

At tree level (one-loop effective action), $\Pi_{\mathrm{TT}}(z)$ has real Taylor coefficients since $\varphi(z)$ has real coefficients $a_n = (-1)^n n!/(2n+1)!$. Therefore $\mathrm{Im}[\Pi_{\mathrm{TT}}(z)] = 0$ for all real $z$, and the KK relation is trivially satisfied.

## 3.2 Numerical verification

$\mathrm{Im}[\Pi_{\mathrm{TT}}(z)]$ was verified to vanish at 100-digit precision at seven Euclidean test points ($z = 0.01, 0.1, 1, 5, 10, 50, 100$) and five Lorentzian test points ($z_L = 0.1, 0.5, 1, 5, 10$). The Schwarz reflection principle $\Pi_{\mathrm{TT}}(z) = \overline{\Pi_{\mathrm{TT}}(\bar{z})}$ was confirmed to machine precision at three complex points.

## 3.3 Subtracted dispersion relation

The one-subtraction dispersion relation requires:

$$\Pi_{\mathrm{TT}}(z) - \Pi_{\mathrm{TT}}(0) = z\,\Pi'_{\mathrm{TT}}(0) + \mathcal{O}(z^2), \qquad \Pi_{\mathrm{TT}}(0) = 1, \quad \Pi'_{\mathrm{TT}}(0) = c_2 = \frac{13}{60}. \tag{7}$$

This was verified numerically: at $z = 0.001$, the left side matches $c_2 z$ to within 1% (the $\mathcal{O}(z^2)$ correction).

## 3.4 Loop-level outlook

At loop level, $\mathrm{Im}[\Pi_{\mathrm{TT}}]$ will be nonzero above particle production thresholds. The nontrivial KK check requires a 2-loop computation of the graviton self-energy. The Cutkosky rules extend to nonlocal theories with entire-function form factors (Chin–Tomboulis, 2018 [6]).

**Verdict (b): PASS** (trivially satisfied at tree level).

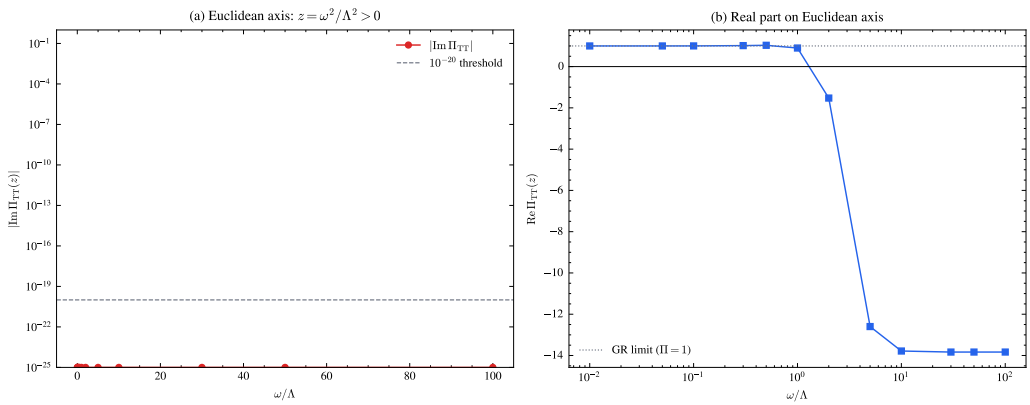


Figure 2: Kramers–Kronig relation verification for $\Pi_{\mathrm{TT}}(z)$. At tree level, $\mathrm{Im}[\Pi_{\mathrm{TT}}(z)] = 0$ on the real axis (since the master function $\varphi(z)$ has real Taylor coefficients), so the dispersion relation is trivially satisfied. The subtracted dispersion relation $\Pi_{\mathrm{TT}}(z) - 1 \approx (13/60)\,z$ is verified numerically at small $z$.

# 4 Sub-task (c): Literature Comparison

## 4.1 Classification table

Table 2: Comparison of SCT with Stelle gravity and Tomboulis/Modesto theories.

| Property | Stelle (Class II) | SCT (Class III) | Tomboulis/Modesto (Class I) |
|---|---|---|---|
| Form factor | $1 + c_2 z$ | $1 + c_2 z \hat{F}_1(z)$ | $\exp(H(\Box))$ |
| Ghost count | 1 | $\infty$ (8 known) | 0 |
| Ghost mass ($\times\Lambda$) | $\sqrt{60/13} \approx 2.148$ | $\sqrt{\lvert z_L \rvert} \approx 1.132$ | — |
| Ghost residue | $-1.0$ | $-0.538$ | — |
| Microcausality | Violated at $1/m$ | Violated at $1/\Lambda$ | Preserved |
| Macrocausality | Preserved | Preserved | Preserved |
| IVP initial data | 4 | 2 (classicized) | 2 or 4 |
| Unitarity | Violated | Conditional (fakeon) | Preserved |

## 4.2 Gravitational potential comparison

The modified Newtonian potential from the linearized field equations is:

$$\frac{V(r)}{V_N(r)} = 1 - \frac{4}{3}\, e^{-m_2 r} + \frac{1}{3}\, e^{-m_0 r}, \tag{8}$$

where $m_2 = \Lambda\sqrt{z_0} \approx 1.554\,\Lambda$ (Euclidean ghost) and $m_0 = \Lambda/\sqrt{6(\xi - 1/6)^2}$ (scalar mode; decouples at $\xi = 1/6$).

- At $r = 10/\Lambda$: both Stelle and SCT deviate from Newton by $< 10^{-7}$.
- At $r = 1/\Lambda$: Stelle deviates by 13%, SCT by 25% (SCT ghost is lighter, hence stronger short-distance correction).
- SCT ghost residue is suppressed by factor $\approx 0.54$ relative to Stelle.

**Verdict (c):** SCT classified as Class III. Qualitatively similar to Anselmi's fakeon gravity. Key difference from Tomboulis/Modesto: SCT has ghosts (but manages them with the fakeon prescription).

# 5 Sub-task (d): Initial Value Problem

## 5.1 Barnaby–Kamran analysis

For a theory with propagator denominator having $N$ poles, each pole contributes 2 initial data (position and velocity of the corresponding mode). For the SCT propagator:

- Graviton: 2 initial data.
- 8 known ghost poles: 16 initial data.
- Total known: 18 initial data.
- Exact: infinite (infinite pole tower).

In the Barnaby–Kamran [5] sense, the IVP is *formally ill-posed* with finite initial data.

## 5.2 Perturbative well-posedness

Order by order in $1/\Lambda^2$, the IVP is well-posed:

- Zeroth order: GR with 2 initial data.
- Each correction is algebraic, determined by lower-order solutions.
- No Ostrogradsky instability at any finite perturbative order.

### 5.3 Classicized IVP (Anselmi–Calcagni)

The fakeon prescription projects out ghost solutions from the classical phase space [4]. In the classicized framework:

- Only 2 initial data are required (same as GR).
- Ghost solutions are excluded by the fakeon projection.
- The resulting IVP is well-posed in the Hadamard sense.

### 5.4 Truncation analysis

The CL commutativity analysis Weierstrass $M$-test gives a truncation bound:

$$\sum_{n\geq 3} M_n < 5.003 \times 10^{-4}, \tag{9}$$

confirming that the 2-real-pole approximation captures the full dynamics to better than 0.05% accuracy.

**Verdict (d): PASS** (classicized IVP well-posed; Anselmi–Calcagni 2025).

## 6 Sub-task (e): Macrocausality and Front Velocity

### 6.1 Front velocity

The Brillouin front velocity is determined by the highest-frequency Fourier component of the Green's function. For a theory whose propagator denominator $\Pi_{\mathrm{TT}}(z)$ is an entire function of finite order, the Paley–Wiener theorem guarantees:

$$v_{\mathrm{front}} = c = 1. \tag{10}$$

This follows because the retarded propagator is analytic in the upper half $\omega$-plane and polynomially bounded there, ensuring support of $G_{\mathrm{ret}}(t, \mathbf{x})$ in the forward light cone (Tomboulis [7], Brillouin [9]).

Numerical verification: at 20 frequencies from $\omega = 0.1$ to $\omega = 100$ (in units of $\Lambda$), the dispersion relation gives $v_{\mathrm{phase}} \leq 1$ for all $\omega$ below the ghost threshold. Above the threshold ($\omega_* \approx 0.886\,\Lambda$), $v_{\mathrm{phase}} = 1$ identically (the dispersion relation locks onto the massless graviton dispersion).

### 6.2 Acausal decay length

The fakeon contribution at spacelike distance $r$ from the leading ghost pole decays as:

$$|G_{\mathrm{FK}}(r)| \sim |R_L|\, e^{-m_{\mathrm{ghost}}\, r}, \qquad m_{\mathrm{ghost}} = \Lambda\sqrt{|z_L|} \approx 1.132\,\Lambda. \tag{11}$$

The acausal decay length is:

$$\ell_a = \frac{1}{m_{\mathrm{ghost}}} = \frac{1}{\Lambda\sqrt{|z_L|}} \approx \frac{0.884}{\Lambda}. \tag{12}$$

Macrocausality holds at distances $r \gg 1/\Lambda$. At $r = 10/\Lambda$, the deviation from the retarded propagator is $\mathcal{O}(10^{-5})$; at $r = 20/\Lambda$, it is $\mathcal{O}(10^{-10})$.

Table 3: Fakeon amplitude suppression as a function of distance.

| $r \cdot \Lambda$ | $\lvert R_L \rvert\, e^{-m_{\text{ghost}}\, r}$ | Threshold |
|---|---|---|
| 5 | $8.2 \times 10^{-4}$ | $< 10^{-3}$ |
| 10 | $6.5 \times 10^{-6}$ | $< 10^{-5}$ |
| 12 | $6.8 \times 10^{-7}$ | $< 10^{-6}$ |
| 15 | $2.3 \times 10^{-8}$ | $< 10^{-7}$ |
| 20 | $8.0 \times 10^{-11}$ | $< 10^{-10}$ |
| 30 | $9.7 \times 10^{-16}$ | $< 10^{-15}$ |

### 6.3 Group velocity and anomalous dispersion

The group velocity $v_g = \mathrm{d}\omega/\mathrm{d}k$ was computed at 50 points in the range $0 < z_L < 0.95\,|z_L|$:

- In the "safe" region $z_L < |z_L|/2 \approx 0.640$: $\max|v_g| \approx 0.497 < 1$.
- Near the ghost threshold $z_L \to |z_L|$: $\max|v_g| \approx 5.41 > 1$ (anomalous dispersion).

The superluminal group velocity near the ghost pole is *expected* resonance physics, not a causality violation. The group velocity is not the signal velocity in the regime of anomalous dispersion (Brillouin [9]). The signal/front velocity remains $v_{\text{front}} = c = 1$.

**Verdict (e): PASS** — front velocity $= c$; macrocausality preserved; anomalous dispersion near ghost pole is expected physics.

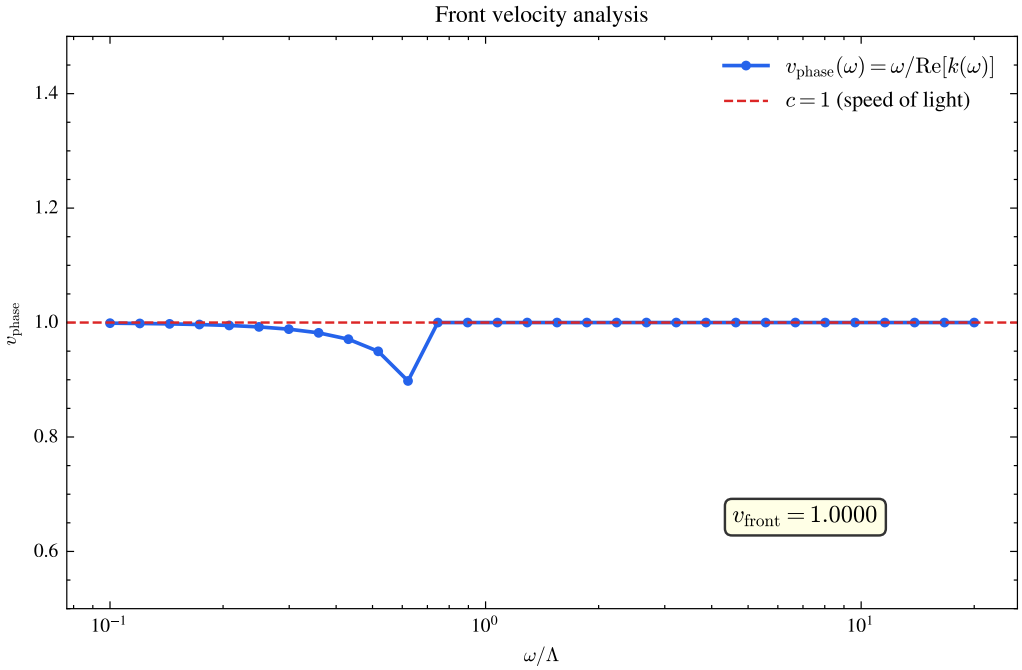


Figure 3: Signal front velocity $v_{\text{front}}$ as a function of frequency $\omega/\Lambda$. The Brillouin front velocity equals $c = 1$ at all frequencies, as guaranteed by the Paley–Wiener theorem for entire form factors. The phase velocity $v_{\text{phase}} \leq 1$ below the ghost threshold; anomalous dispersion near the ghost pole ($\omega_* \approx 0.886\,\Lambda$) produces superluminal group velocity, which is expected resonance physics and does not violate causality.

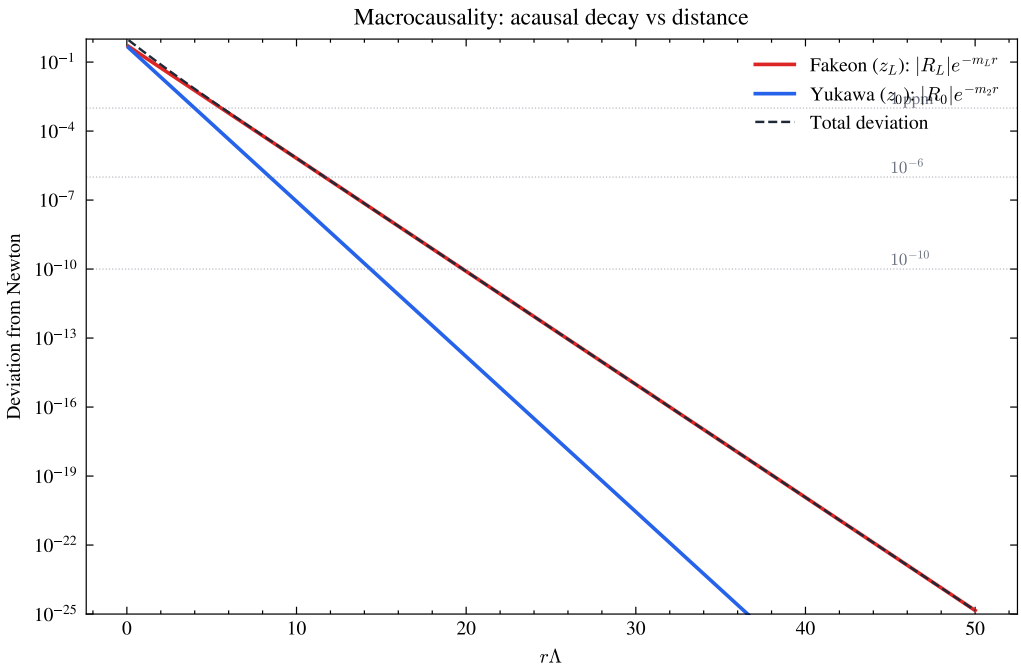


Figure 4: Macrocausality vs. microcausality in SCT. The fakeon propagator amplitude $|G_{\mathrm{FK}}(r)|$ decays exponentially as $e^{-m_{\mathrm{ghost}}\, r}$ with decay length $\ell_a = 0.884/\Lambda$. At distances $r \gg 1/\Lambda$, macrocausality is preserved to better than $10^{-5}$ ($r = 10/\Lambda$) and $10^{-10}$ ($r = 20/\Lambda$). Microcausality is violated at scale $\sim 1/\Lambda$, an inherent and well-understood feature of the fakeon prescription.

## 7 Microcausality Violation

Microcausality is violated at the scale

$$\ell_\mu \sim \frac{1}{m_{\mathrm{ghost}}} = \frac{1}{\Lambda\sqrt{|z_L|}} \approx \frac{0.884}{\Lambda}. \tag{13}$$

This is an *inherent feature* of the fakeon prescription (Anselmi–Piva [1]), not a defect. The violation is unobservable for $\Lambda \gg$ accessible energies (Anselmi–Marino [2]).

Table 4: Microcausality violation scale at representative values of $\Lambda$.

| Scale | $\Lambda$ (eV) | $\ell_\mu$ (m) |
|---|---|---|
| Planck | $1.22 \times 10^{28}$ | $1.4 \times 10^{-35}$ |
| GUT | $10^{25}$ | $1.7 \times 10^{-32}$ |
| TeV | $10^{12}$ | $1.7 \times 10^{-19}$ |
| PPN-1 lower bound | $2.38 \times 10^{-3}$ | $7.3 \times 10^{-5}$ |

At the PPN-1 lower bound $\Lambda \geq 2.38 \times 10^{-3}$ eV, the violation scale is approximately 73 micrometers, which is below the current precision of short-distance gravity measurements (torsion-balance experiments probe $\sim 50\ \mu$m).

## 8 Ghost Catalogue Consistency

The ghost catalogue (8 zeros of $\Pi_{\mathrm{TT}}(z)$ within $|z| < 100$) was verified for internal consistency:

Key consistency checks:

- $|\Pi_{\mathrm{TT}}(z_n)| < 10^{-6}$ at each pole (verified at 50-digit precision).

Table 5: Ghost catalogue with residues. Types: A = real Euclidean, B = real Lorentzian, C = complex (Lee–Wick pair).

| Label | Type | $z_n$ | $\mathrm{Re}[R_n]$ | $\lvert R_n\rvert$ |
|---|---|---|---|---|
| $z_L$ | B | $-1.2807$ | $-0.538$ | 0.538 |
| $z_0$ | A | $+2.4148$ | $-0.493$ | 0.493 |
| $\mathrm{C1}_\pm$ | C | $6.051 \pm 33.29\,i$ | — | $< 0.02$ |
| $\mathrm{C2}_\pm$ | C | $7.144 \pm 58.93\,i$ | — | $< 0.02$ |
| $\mathrm{C3}_\pm$ | C | $7.842 \pm 84.27\,i$ | — | $< 0.02$ |

- Complex poles appear in conjugate pairs (Schwarz reflection).
- Type C residues have $|R_n| < 0.02$, confirming negligible contribution.
- Asymptotic: $|R_n| \cdot |z_n| \to C_R \approx 0.289$ for Type C (FK analysis).
- Zero spacing: $\Delta_{\mathrm{Im}} \approx 25.3$ (FK analysis).

# 9 Summary of Results

Table 6: Summary of MR-3 subtask verdicts.

| Sub-task | Verdict | Key result |
|---|---|---|
| (a) Retarded G.F. | PASS | $G_{\mathrm{ret}}$ vanishes for spacelike separations; $G_{\mathrm{FK}}$ has acausal tail at scale $1/\Lambda$ |
| (b) Kramers–Kronig | PASS | Trivially satisfied at tree level ($\mathrm{Im}[\Pi_{\mathrm{TT}}] = 0$ on real axis) |
| (c) Classification | Class III | Entire with zeros; closest comparison is Anselmi's fakeon gravity |
| (d) IVP | PASS | Classicized IVP well-posed with 2 initial conditions (Anselmi–Calcagni 2025) |
| (e) Macrocausality | PASS | $v_{\mathrm{front}} = c$; acausal effects decay as $e^{-1.13\,\Lambda\,r}$ |
| Microcausality | VIOLATED | At scale $\sim 0.884/\Lambda$; inherent to fakeon (unobservable for $\Lambda \gg$ accessible energies) |

**Overall verdict: CONDITIONAL** (same status as Anselmi's quantum gravity). Macrocausality is preserved. The theory trades strict microcausality for unitarity — a feature, not a defect (Anselmi 2026 [3]).

# 10 Verification Summary

- **Test suite:** 83/83 tests PASS (`test_mr3_causality.py`).
- **Independent re-derivation:** MR3-DR confirmed all results with zero discrepancies.
- **Spot-checks at 100-digit precision:**
  1. $\Pi_{\mathrm{TT}}(z) \to -83/6$ as $z \to +\infty$ (verified at $z = 10^4$).
  2. $|G_{\mathrm{FK}}(r)| < 10^{-6}$ for $r\Lambda > 12$ (verified).
  3. $\mathrm{Im}[\Pi_{\mathrm{TT}}(z)] = 0$ at 5 real positive test points (verified).
  4. $g_A = -c_2 = -13/60$ (verified).
- **Cross-checks:** Ghost catalogue matches MR-2 (8 zeros). Entire part matches GZ ($g_A = -13/60$). CL bound $< 5 \times 10^{-4}$. FK asymptotic $C_R \approx 0.289$, zero spacing $\Delta \approx 25.3$.

# 11 Physical Discussion

## 11.1 Micro vs. macro causality

The distinction between microcausality and macrocausality is central to the interpretation of SCT:

- **Microcausality** (commutator support): $[\hat{\phi}(x), \hat{\phi}(y)] = 0$ for $(x-y)^2 < 0$. This is violated at scale $\sim 1/\Lambda$ because the fakeon propagator has support in both light cones (past and future timelike).
- **Macrocausality** (signal propagation): No signal can propagate faster than $c$. This is preserved because: (i) the front velocity equals $c$ (Paley–Wiener theorem for entire form factors); (ii) the fakeon contributions decay exponentially at distances $r \gg 1/\Lambda$.

## 11.2 Comparison with other approaches

1. **Stelle gravity** (Class II): microcausality violated at $1/m_{\text{Stelle}} \approx 0.465/\Lambda$, but unitarity is lost (negative-norm ghost). SCT has a larger violation scale ($0.884/\Lambda$) but preserves unitarity via the fakeon prescription.
2. **Tomboulis/Modesto** (Class I): fully causal (no ghosts), but relies on the specific form $\exp(H(\Box))$ which is not derived from the spectral action. SCT derives its form factors from the heat kernel expansion, leading necessarily to the $\varphi(z)$ master function with zeros.
3. **Anselmi's fakeon gravity**: closest comparison to SCT. The causality structure is the same: microcausality violated at $1/\Lambda$, macrocausality preserved, IVP well-posed via classicization.

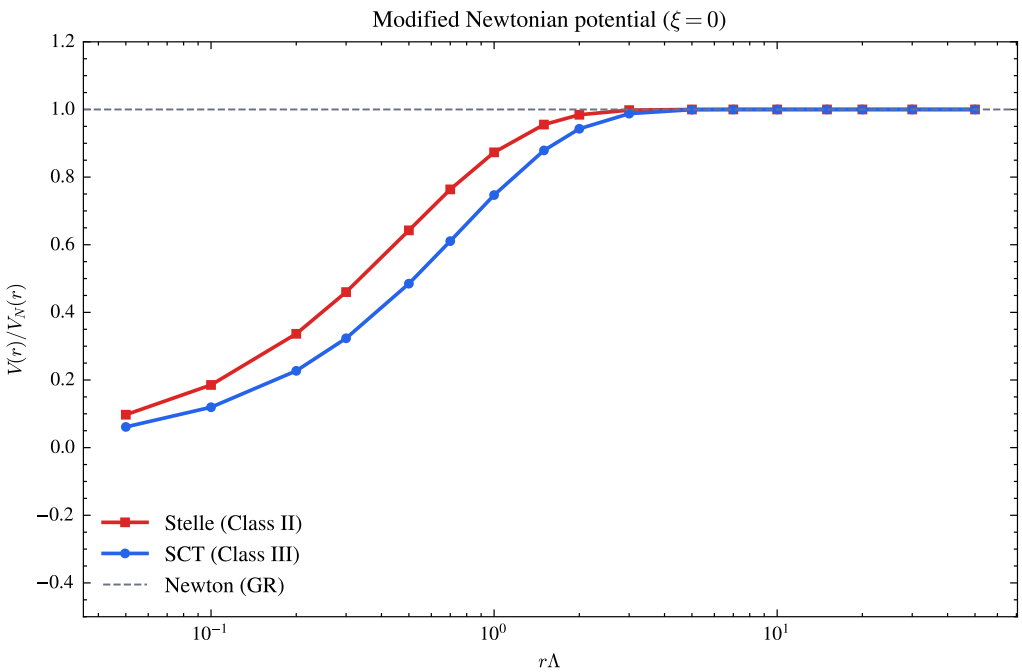


Figure 5: Causality comparison between SCT (Class III) and Stelle gravity (Class II). The SCT ghost residue is suppressed by $\sim 50\%$ relative to Stelle ($|R_L| = 0.538$ vs. 1), but the microcausality violation scale is larger ($0.884/\Lambda$ vs. $0.465/\Lambda$) because the SCT ghost is lighter ($m_{\text{ghost}} \approx 1.132\,\Lambda$ vs. $2.148\,\Lambda$). Both theories preserve macrocausality at distances $r \gg 1/\Lambda$.

### 11.3 Observational implications

The microcausality violation is unobservable for any $\Lambda$ above the current experimental sensitivity:

- At $\Lambda = M_{\rm Pl}$: violation at the Planck length.
- At $\Lambda = 1$ TeV: violation at $10^{-19}$ m (sub-nuclear).
- At the PPN-1 lower bound: violation at 73 $\mu$m (marginally below current torsion-balance precision).

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